



Deprescribing for People with Dementia: A Roadmap

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Abstract

People with dementia (PWD) are frequently exposed to polypharmacy and potentially inappropriate medication use, in which the risks of medication use outweigh the benefits or the medication is not aligned with treatment goals. Appropriate deprescribing of unnecessary or potentially inappropriate medications is essential to high-quality care for PWD, to avoid iatrogenic harm and improve health and well-being for patients and their care partners. In this article, we review the risks of polypharmacy in PWD and evidence for the safety and efficacy of deprescribing in this population. Building off existing deprescribing frameworks for older adults with multimorbidity and limited life expectancy, we provide a roadmap for deprescribing in PWD that addresses the unique challenges of living dementia, including the variable disease course, high prevalence of distressing behavioral symptoms, and central role of care partners. The steps include: (1) identify potential targets for deprescribing by eliciting medication-related goals and considering tradeoffs, (2) develop a tapering plan, (3) complete additional actions that are necessary before deprescribing, and (4) provide close follow-up. Lastly, we provide evidence-based strategies for communicating with patients and their care partners about deprescribing, adapted from the FRAME tool.

Key Points

People with dementia frequently experience polypharmacy, in some cases exposing them to iatrogenic harm with little likelihood of benefit.

A structured approach to deprescribing is needed to safely reduce unnecessary polypharmacy and provide high-quality care to people with dementia.

This article proposes an adapted framework for deprescribing specific to people with dementia, along with evidence-based strategies for communicating with patients and their care partners.

1 Introduction

Globally, 57 million people are living with dementia, with nearly 10 million new cases diagnosed each year [1]. In the USA alone, the number of people with Alzheimer's disease and related dementias (ADRD) will reach 14 million by 2060 [2]. People with dementia (PWD) use more medications and have more complex medication regimens than people without dementia. A recent systematic review reported that the pooled prevalence of polypharmacy (defined as five or more medicines) among PWD was 62% [3]. Polypharmacy increases the risk of potentially inappropriate medication (PIM) use, in which the risks of medication use outweigh the benefits or the medication is not aligned with treatment goals [4]. Up to 56% of PWD are taking at least one PIM [5], and over half of nursing home residents with severe ADRD are prescribed at least one medication with uncertain benefits for such patients, such as statins and cholinesterase inhibitors.¹ This is owing to the presence of multiple comorbidities, such as heart disease, hypertension and diabetes, and geriatric conditions such as urinary incontinence and chronic pain. In addition, medications are often prescribed to treat challenging behavioral and psychological symptoms associated with ADRD, such as sleep disturbance, depression, anxiety, agitation, and psychosis.

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The risks of polypharmacy and PIM use, including drug interactions, treatment burden, cognitive adverse effects, falls and fractures, hospitalizations, and mortality, are exacerbated in PWD [6–11]. Pharmacokinetic changes that occur with aging, including reduced metabolism and clearance of medications, increase the risk of adverse drug effects (Fig. 1) [12, 13]. Metabolic changes common among PWD, such as alterations in body composition, a high prevalence of renal impairment, and increased permeability of the blood brain barrier, also increase the risk of medication harms, such as falls, delirium, and mortality [14–16]. As cognitive impairment progresses, the risks of polypharmacy in PWD are heightened by high rates of dysphagia [3], difficulty following directions, and taking medications appropriately, and inability to recognize symptoms that may be owing to adverse effects of medications. As a result of these challenges, hospital admissions owing to adverse drug events are three times higher in PWD as compared with people without dementia (14 versus 4%) [17]. Managing complex medication regimens is also a major source of strain for family and others who are involved in helping the PWD (“care partners”) owing to the time and stress associated with administering medications and monitoring for side effects, fear of making mistakes, and lack of training [18]. PWD are also exposed to PIM in long-term care facilities, where behavioral disturbances can be especially challenging to manage in a group setting and medications are administered by staff who may not be trained in non-pharmacological approaches [19].

A principle of person-centered care is that people should only be on the medications that will support their goals—while avoiding those that are likely to be ineffective or harmful [20]. For this reason, appropriate deprescribing of unnecessary or potentially inappropriate medications is essential to high-quality dementia care. Deprescribing is a proactive process designed to reduce unnecessary and potentially harmful medication use and improve health and well-being for older adults, particularly those with multiple chronic conditions and serious, life-limiting diagnoses such as dementia. National surveys of older adults have revealed strong support for deprescribing [21].

The challenge of deprescribing for PWD is illustrated by the story of Ms. R, an 89 year old woman with Alzheimer’s disease and multiple chronic conditions who presents to the primary care clinic to establish care (Table 1). She is accompanied by her daughter, who lives with her and is her primary care partner. Ms. R’s daughter is concerned that her mother is overly sedated and has recurrent falls. Ms. R is unable to answer questions or to stand without assistance. Her score on the Mini-Mental State Exam is 3 out of 30, consistent with advanced dementia. In addition to Alzheimer’s disease, she has type 2 diabetes, chronic obstructive pulmonary disease, hypertension, osteoporosis, chronic low back pain, urinary incontinence, sleep disturbance, and

anxiety. She has fallen four times in the past year. Her daughter administers her 14 daily medications, which require up to 25 doses per day. Her regimen includes high-risk medications such as insulin, a sedative-hypnotic, and a strongly anticholinergic overactive bladder medicine. How can Ms. R’s new primary care clinician identify suitable targets for medication rationalization, discuss decisions about stopping and starting medications with her daughter, and implement deprescribing for Ms. R?

In this article, we summarize the problem of medication use in PWD, evidence for the safety and efficacy of deprescribing in PWD, and challenges to deprescribing in PWD. We provide a adapted roadmap for deprescribing specific to older adults with ADRD, and strategies for communicating with patients and their care partners about deprescribing that can be used together with the roadmap. While our focus is primary care, as it is the site in which most older adults, including PWD, receive their medical care [22], the same principles can be applied to any older adult with ADRD, regardless of setting.

2 Evidence of Efficacy and Safety of Deprescribing

Few deprescribing interventions in primary care have been specific to people with mild cognitive impairment (MCI) or dementia [23–25]. Studies have primarily focused on deprescribing of either psychotropic medications or PIM, with some addressing the entire medication regimen [23]. Interventions have included medication reviews led by geriatricians, pharmacists, interdisciplinary teams, or computer algorithms that triggered alerts to professionals. Some studies involved educational interventions aimed at patients, care partners, and/or healthcare professionals. Inappropriate medications were identified through tools such as the Beers criteria, the Screening Tool of Older Person’s Prescriptions (STOPP), anticholinergic burden scores, or intervention-specific tools. Other deprescribing interventions have focused on nervous system (CNS)-active medications, which are especially high-risk for PWD [26]. For example, the Veterans Health Administration (VHA) and the US Centers for Medicare and Medicaid Services have attempted to reduce off-label use of antipsychotics in PWD in nursing homes through training of nursing home staff, publicly-reported measures of antipsychotic drug use, and psychopharmacology quality improvement initiatives. While antipsychotic and anxiolytic prescribing in VHA nursing homes declined, prescribing of alternative psychotropic medicines, such as gabapentin, increased. This use is in spite of less proven benefit and comparable risks [27].

In general, studies of deprescribing interventions in people with MCI or dementia have shown a positive effect on

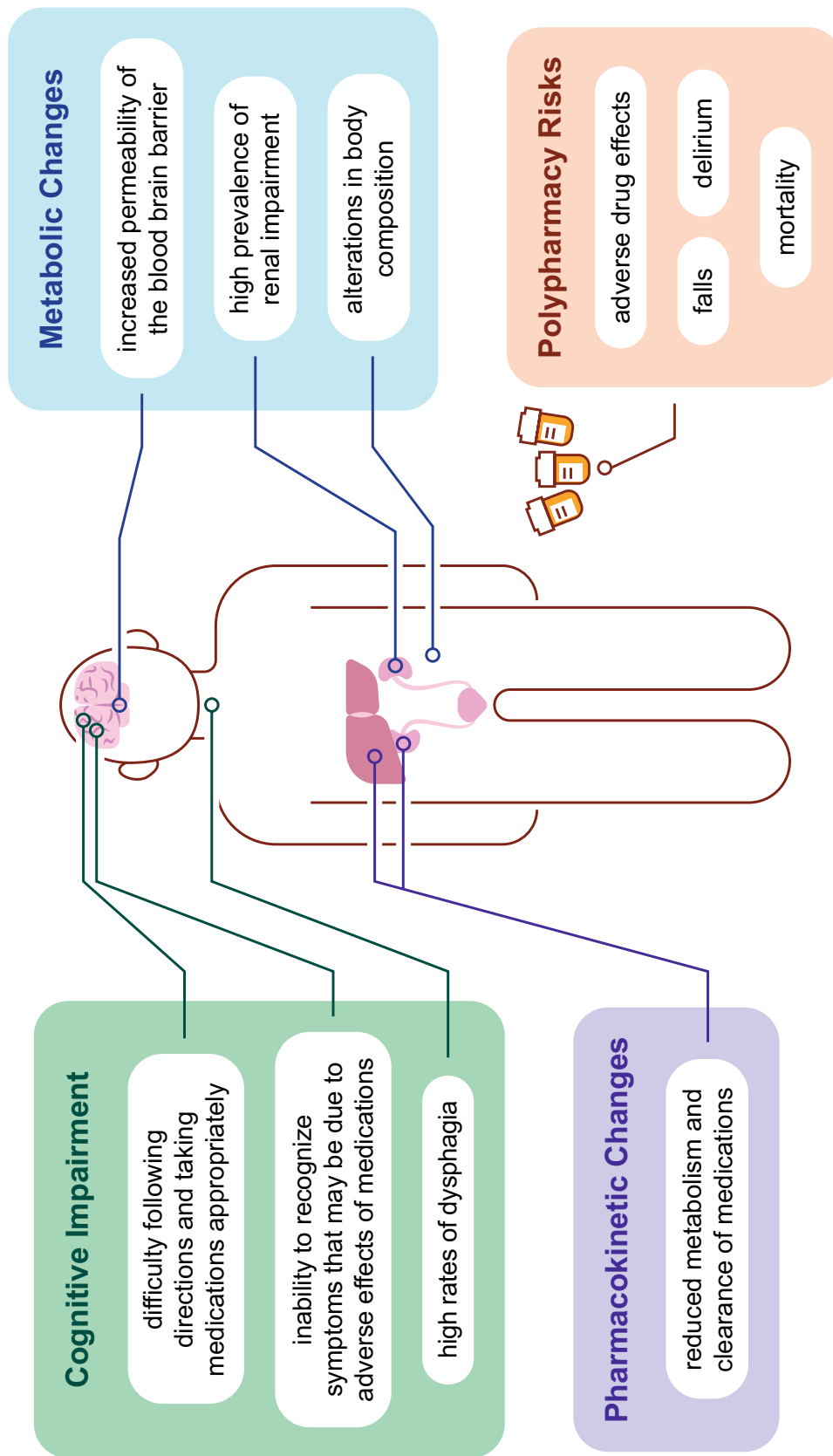


Fig. 1 Factors that increase the risk of adverse drug effects for PWD

Table 1 Ms. R, an 89-year-old hypothetical patient

Living arrangement: lives with daughter, who is her primary care partner
Daughter's priorities: reduce her mom's sedation, prevent falls
Functional status: dependent for instrumental and most basic daily activities
Medical problems:
Advanced Alzheimer's disease
Type 2 diabetes mellitus
Chronic obstructive pulmonary disease
Hypertension
Osteoporosis
Chronic low back pain
Urinary incontinence
Anxiety
Medications: 14 daily (up to 25 doses per day); multiple high-risk

medication-related outcomes, such as reducing the total number medications and PIM. However, a notable gap in the literature is that few studies have reported on clinical outcomes, such as cognition, falls, fractures, delirium, behavioral and psychological symptoms of dementia (BPSD), and quality of life, and the findings from those that have are mixed [28]. The most frequently measured clinical outcome was BPSD, assessed mainly in long-term care settings. Importantly, the findings of a recent systematic review of deprescribing studies in PWD show that a decrease in psychotropic medication had either no effect or resulted in an improvement in BPSD [28].

The growing body of evidence described above provides support for deprescribing interventions that involve patient-directed education rather than interventions focused only on provider prescribing behaviour [29–32]. For example, a “low-touch” direct-to-patient deprescribing intervention, EMPOWER, was particularly effective for reducing chronic benzodiazepine use in older adults, including those with MCI [32, 33]. Patients randomized to the intervention group received a personalized brochure by mail that encouraged them to discuss deprescribing with their physician and/or pharmacist. The intervention resulted in 27% of the intervention group discontinuing benzodiazepine use by 6 months, compared with 5% of the control group [32].

More recent studies, OPTIMIZE [23] and DPREScribe-AD [24], have attempted to adapt the low-touch EMPOWER approach to PWD in the USA with less success. These studies, in which PWD received mailed educational brochures and their primary care clinicians received deprescribing information, did not significantly reduce the number of long-term or potentially inappropriate medications [23, 24]. One reason these trials had null effects in contrast to EMPOWER may be that PWD, their care partners and clinicians require more in-depth conversations about risks and benefits of medications, particularly as the PWD's cognitive and functional status declines. Another may be that certain PIM may be easier to deprescribe, or that interventions targeting the

entire regimen may be more successful in patients with the highest medication counts [23].

Overall, research suggests that deprescribing among older adults, including those with ADRD, is safe. This is especially true when it is carried out through a patient-centered, structured process involving tapering and close monitoring [34–38]. Studies have shown no adverse impacts of deprescribing on hospitalization or mortality, and some systematic reviews have suggested a reduction in all-cause mortality with deprescribing [39, 40]. Of note, while the relative lack of evidence for the effectiveness and safety of deprescribing interventions hinders translation into the real-world, evidence is also lacking for the benefits of *prescribing* multiple medications in older adults, especially those with multimorbidity, cognitive impairment, and frailty [41].

3 Deprescribing in PWD: Challenges and Opportunities

There are several challenges unique to deprescribing in PWD. The first is the nature of the condition itself. Given the prolonged and variable time course of dementia, determining whether a medication to prevent or treat a coexisting condition is likely to result in benefits during the person's remaining lifespan is difficult. PWD often have multiple comorbidities and fluctuations in health status, making it difficult to sustain deprescribing over longer intervals. Optimizing prescribing for PWD may be further complicated by impaired decision making capacity, distressing BPSD, care partner strain, and the difficulty of establishing goals of care [42]. In addition, qualitative research has shown that patients and care partners may perceive a clinician's recommendation to deprescribe as a withdrawal of care [43, 44]. Finally, it can be challenging to show clinical benefits of deprescribing in PWD due to the short duration of follow-up in most trials—particularly with respect to outcomes such as cognition, which may take longer to become clinically observable.

Clinician and health system barriers include guidelines and quality metrics that focus on single conditions and typically promote medication initiation, incomplete medical histories, the involvement of multiple prescribers in a patient's care, time pressures during clinical encounters, avoidance of negative consequences, and a culture that tends to over-emphasize the benefits of medication use and downplay the risks [42]. Lastly, the difficulty of implementing practical, effective nonpharmacologic therapies and the lack of access to supportive services in the community for PWD and care partners makes it difficult for clinicians to avoid prescribing high-risk medications such as antipsychotics and sedative-hypnotics to treat BPSD [45].

Despite these challenges, there are unique opportunities to enhance deprescribing among PWD, particularly in primary care. First, is the Centers for Medicare and Medicaid Services Guiding an Improved Dementia Experience (GUIDE) model. This 8-year national program provides a payment mechanism for health care systems to provide comprehensive dementia care, which notably includes deprescribing as well as care partner supports and training to address BPSD and caregiver strain through non-pharmacologic approaches [46, 47]. Second, primary care has a unique vantage point to address deprescribing. Primary care clinicians often have a longitudinal relationship with patients and care partners, which can facilitate trust [48]. Primary care takes a holistic view of the PWD and all of their conditions, which may allow them to better weigh the risks of benefits of deprescribing.

4 A Roadmap for Deprescribing in PWD

Numerous frameworks have been developed for deprescribing in older adults and in people nearing the end of life. The most-cited deprescribing protocol, by Scott et al., involves five steps: (1) identify all medications the patient is taking and the indications for each one, (2) evaluate the overall risk of medication-related harm in light of the patient's unique circumstances to determine the appropriate intensity of deprescribing, (3) consider each medication's potential for current or future benefits compared with its potential current or future harms or burdens, (4) prioritize deprescribing medications with the least favorable benefit-risk ratio and the lowest risk of withdrawal symptoms or disease relapse, and (5) implement a deprescribing plan and monitor the patient closely [34].

Another framework, the Deprescribing Rainbow, is intended to be used in conjunction with other deprescribing schemas, such as the one by Scott et al. The Rainbow highlights five deprescribing determinants that should be considered in deprescribing for older adults with multimorbidity [49]—clinical, psychological, social, financial, and

physical. It places the older adult at the center of the rainbow to illustrate the importance of using patient-centered communication to account for the complex patient context and identify which determinants are a priority for the patient. Factors such as the patient's remaining life expectancy and the time to benefit of medications are considered.

Finally, Holmes et al. adapted previous frameworks to describe a process for medication prescribing and deprescribing in patients near the end of life [50]. This model has four components that together determine the appropriateness of a given medication for a patient who is late in life: remaining life expectancy, time until benefit, goals of care, and treatment targets, such as palliation of symptoms versus primary and secondary prevention. For example, some patients may desire a more palliative model in which their only medications would address bothersome symptoms. By contrast, other patients may prioritize life prolongation or prevention of morbidity and mortality.

Informed by our previous research, our research team has identified and adapted the elements of existing deprescribing frameworks that are most relevant to PWD and created a roadmap to guide deprescribing for this population [25, 51]. Key considerations include: (1) deprescribing for PWD needs to account for the prolonged and variable trajectory of dementia. The average life expectancy of a person with AD/DR is 8–10 years, though depending on the age of diagnosis, some people with AD/DR may live for 15–20 years. (2) The majority of PWD will eventually experience distressing symptoms of dementia, such as mood disorders, sleep disturbance, psychosis, and agitation. Given the dearth of accessible, nonpharmacologic resources to cope with these resources, many family care partners currently rely on PIM, such as sedative-hypnotics, to enable the PWD to continue living at home, despite the lack of evidence of effectiveness and safety for these medications. (3) As dementia progresses, deprescribing conversations will increasingly involve care partners, who have unique informational and decisional needs that must be addressed.

The adapted framework for deprescribing in PWD is described below and presented in Fig. 2:

1) Identify potential targets for deprescribing. This involves two steps, which are important for all patients but especially for PWD, given the progressive, debilitating nature of dementia and its variable time course:

1a) Elicit medication-related goals and priorities: asking PWD and care partners “what matters most” to them is an essential component of providing care that is patient-centered and addresses older adults' physical and emotional needs [52]. Identifying and prioritizing older adults' specific health outcome goals is particularly important for PWD, who are especially vulnerable to experiencing harm as a result of unwanted care. Different members of the healthcare team, including pharmacists, nurses and social workers, can

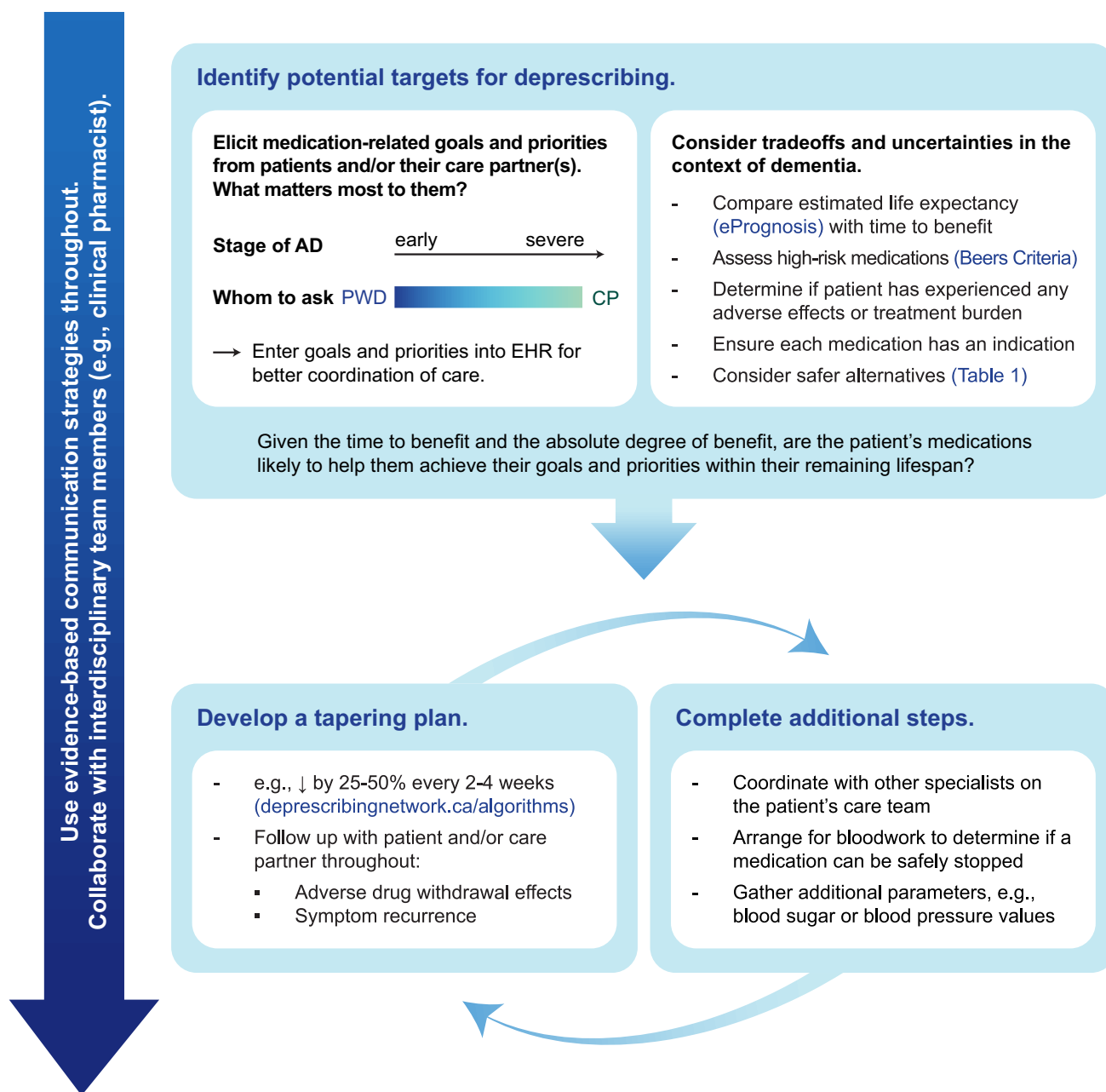


Fig. 2 Adapted roadmap for deprescribing in PWD. PWD people with dementia, CP care partner

be encouraged to ask “what matters most,” and to document the response in the patient’s electronic health record to inform clinical decision-making. In the early to moderate stages of AD, the patient may be able to participate in the discussion. As dementia progresses, the role of the care partner will become more central in such conversations.

There are several ways to frame the question about goals and priorities. For example:

- “If we changed your medicines, what do you wish we could help with?”
- “Are there any activities you would like to perform that your health—or medications—currently prevent you from doing?”
- “What are your most important goals for your health care over the next 6 months to a year?”
- What are you hoping to achieve with your medicines?
- Where do you find the most joy in life?

Once the patient's health outcome goals and priorities have been identified, the clinician should assess whether the patient's medications are likely to help them achieve what matters most to them, as described in section 1b. In Ms. R's case, for example, her daughter's priorities are to lessen her mother's sedation to allow her to engage with family and reduce falls. Her overarching goal is to support her mother's quality of life and focus on comfort rather than life prolongation. These goals serve as guideposts for decisions about Ms. R's medications. A barrier is that health systems currently lack standardized approaches to collect and document patient goals and priorities in the electronic medical record so that they are visible to all members of the healthcare team and can inform goal-aligned care [53].

1b) Consider tradeoffs and uncertainties: the next step is to consider the risks, benefits, and uncertainties of each medication, specifically in light of the patient's dementia and overall health. In other words, is the medication likely to help this patient achieve their priorities, and if so, by how much? A recent qualitative interview study with care partners of patients with dementia found that they preferred when clinicians initiated deprescribing conversations with an individualized risk versus benefit discussion that specifically addressed the patient's functional and cognitive status. They also valued knowing a medication's potential impact on the patient's dementia [54].

One challenge is the lack of evidence of deprescribing outcomes, especially for PWD. Another challenge that time to benefit information is not always available or straightforward for PWD, who are often excluded from clinical trials. Finally, time to benefit does not consider

that severe cognitive and functional impairment may make certain interventions less desirable in the advanced stages of dementia. When discussing deprescribing, clinicians caring for PWD should share the best available evidence, acknowledge its limitations, and explain that multiple reasonable options exist, with the choice depending on the patient's and care partner's preferences [50]. Several questions and resources can guide this process:

- **Is the medication likely to help this patient within their remaining lifespan?** If so, is it possible to estimate the absolute degree of benefit or is there significant uncertainty based on the available data? ePrognosis (<https://eprognosis.ucsf.edu/>) is an online repository of life expectancy calculators based on published geriatric prognostic indices. Clinicians can use ePrognosis to obtain evidence-based information on patients' prognosis to inform clinical decision-making. It includes a prognostic calculator designed specifically for patients with dementia. Returning to Ms. R from the vignette in the Introduction, her remaining life expectancy based on ePrognosis is 2.9 years. This suggests that she is not likely to benefit from intensive glycemic control (hemoglobin A1c < 7.0%), which has a time to benefit of 8–10 years for microvascular complications. Because of her dementia, Ms. R is at greater risk of treatment-related harms, including hypoglycemia and falls. This suggests that Ms. R's glycemic target should be de-intensified and that one or more of her diabetes medications can be stopped.

Table 2 Alternatives for commonly-used high-risk medication classes in PWD

Symptom/condition	Alternative
Chronic pain	Physical activity: many senior centers and adult day programs have exercise classes that are designed for PWD. Area Agencies on Aging are a good starting point for learning about programs for PWD, including exercise classes Topical analgesia, such as topical diclofenac, lidocaine, capsaicin and corticosteroids Injections for pain (e.g., corticosteroid injections) Heat and cold therapy, massage, and positioning may also be beneficial
Anxiety and depression	Psychotherapy or cognitive behavioral therapy (CBT; in earlier stages of dementia) Structured, supervised activity in more advanced stages (e.g., adult day programs for PWD)
Urinary incontinence	Lifestyle interventions (e.g., scheduled toileting, limiting bladder irritants)
Agitation	Structured activity (e.g., adult day program) Music
Sleep disturbance	Sleep hygiene (e.g., structured daytime activities, relaxing bedtime routine, comfortable sleep environment) Unfortunately, although CBT is the first-line treatment for insomnia in older adults, there is limited evidence for CBT in PWD, and sedative-hypnotics are not recommended as they carry considerable risks in this population [62]

PWD people with dementia, CBT cognitive behavioral therapy

- **Which medications are especially high-risk?** The Beers Criteria is a comprehensive list of medications that older adults should avoid or use with caution because evidence suggests that risks outweigh benefits for most individuals over the age of 65 years. For PWD, these include strongly anticholinergic medicines, benzodiazepines, sedative-hypnotics, and anti-psychotics, which can exacerbate cognitive symptoms and cause delirium. CNS-active polypharmacy, which refers to the concurrent use of three or more CNS-active medications, is common and especially risky among PWD, with most of these prescriptions stemming from primary care [26, 55]. In addition, medications such as insulin and anticoagulants require precise management, which people with cognitive impairment may have difficulty implementing. Ms. R is taking several medicines that are high-risk for PWD.
- **Has the patient experienced adverse effects or treatment burden?** Clinicians should ask specifically about geriatric syndromes or symptoms that are often overlooked as medication-related, such as delirium, dizziness, falls, sedation, urinary incontinence, and poor appetite or weight loss. Ms. R has experienced some of these symptoms, which may be caused or exacerbated by medications. For example, xerostomia, nausea, or constipation can contribute to poor appetite and weight loss, which are already common problems as dementia progresses. Deprescribing can frequently alleviate these symptoms and improve quality of life for people with dementia.
- **Which medications have an indication?** Clinicians should consider why and when a medication was started, whether the medication is truly helping with that symptom, and whether the drug was prescribed to treat adverse effects of another drug (the prescribing cascade). For instance, bladder antimuscarinics are often prescribed to PWD, including Ms. R, despite recommendations to avoid them due to the risk of anticholinergic-induced adverse effects, including worsening cognition and constipation, which may be difficult to manage in PWD. Since urinary incontinence in PWD is multifactorial, and cognitive impairment is often a major contributor, bladder antimuscarinics may not effectively reduce incontinence episodes in a way that is meaningful to the patient or care partner or worth the adverse effects. In addition, studies have shown that PWD who are prescribed cholinesterase inhibitors for dementia have an increased risk of subsequently receiving a bladder antimuscarinic to treat urinary incontinence—an example of a prescribing cascade. Clinicians should be alert to this potentially reversible medication-related adverse event, rather than misinterpreting incontinence as an inescapable symptom of dementia.
- **Are there safer alternatives (including non-pharmacologic)?** Many pharmacologic therapies that are high-

risk for PWD can be replaced with equally effective, safer alternatives (listed in Table 2). However, a barrier is that nonpharmacologic approaches may be challenging to implement and community support remains inaccessible for many PWD and their families:

2) Develop a tapering plan. Although there is limited evidence on how to discontinue most medication classes, especially for PWD, it is generally safest to taper the dose over weeks to months, depending on how long the patient has been taking the drug. This is especially true for certain drug classes, such as benzodiazepines and nonbenzodiazepine receptor agonists, which should not be stopped abruptly because they can cause psychological and physiological dependence. A tapering schedule and patient education materials used in the EMPOWER and D-PRESCRIBE trials are available at <https://www.deprescribingnetwork.ca/patient-handouts>. Other medication classes, such as statins, can be immediately discontinued. It can sometimes be useful to taper symptomatic medications such as proton pump inhibitors to reduce the chance of adverse drug withdrawal effects or symptom recurrence, reassure patients and care partners, and increase their willingness to try deprescribing. A common approach is to reduce the dose by 25–50% every 2–4 weeks, asking the patient or care partner to check in throughout the process.

3) Complete any additional actions that are necessary before deprescribing. Clinicians often report that they are hesitant to stop medications started by other prescribers [56]. It will often be necessary to coordinate with specialists. For example, clinical practice guidelines recommend that patients with heart failure take medications from four different classes; additional therapies may be prescribed for some patients. This complicated regimen is not feasible or goal-aligned for many PWD as their cognitive and functional status deteriorates [57]. In such instances, primary care clinicians should communicate with the patient's cardiologist that the patient's goals and priorities may no longer be aligned with the guidelines in light of their dementia and work together to ensure that deprescribing is safe for the patient [58].

Other times, while the overall goal may be de-intensification of therapies as dementia progresses, it may be reassuring to patients and care partners to arrange for the patient to have bloodwork to determine whether a medication can safely be stopped, or to gather additional clinical parameters (i.e., blood sugar or blood pressure values). This can be especially helpful and may prevent adverse drug withdrawal events while tapering diabetes medications or an anti-hypertensive drugs.

4) Provide close follow up throughout the deprescribing process. In qualitative interview studies, PWD and their

Table 3 Communication strategies for initiating deprescribing conversations with PWD and care partners [60]

Communication strategy	Sample phrases and rationale
F: fortify trust (establish rapport/validate)	Validate patient and care partner lived experience and priorities: “Your biggest concern is what happens if the agitation gets worse and kind of is unmanageable...I certainly can understand that.” [59] Next, explain the rationale for deprescribing while tying it back to their priorities. This is important to assure patients and care partners that deprescribing is a positive step intended to improve well-being and quality of life, not a withdrawal of care [43]. For example, “My role is to help ensure [your] medicines are truly helping [you] achieve what matters most to...you about [your] health” Clinicians should reassure patients and care partners that deprescribing will occur gradually, that they will continue to address the patient’s priorities, such as bothersome symptoms, and will remain flexible to make adjustments as needed [43]
R: recognize willingness to deprescribe (or barriers to deprescribing)	Begin by asking questions to assess patient and care partner readiness to consider deprescribing, such as “How do you feel about your medicines?” and “Do you feel your medications are inconvenient to take? Which ones?” Listen empathetically to expressed barriers and empower self-efficacy by noting incremental changes
A: align deprescribing with goals and priorities	Qualitative research has shown that older adults’ willingness to deprescribe is associated with their health outcome priorities. For example, many older adults prioritized taking fewer medicines and avoiding future side effects over “feeling better now.” [63, 64] Deprescribing can be aligned with goals of care through statements such as, “Thank you for telling me that you want to spend more time with your grandchildren. I want to discontinue your zolpidem as this medication may be causing the mental foggyiness you have been experiencing, which makes it difficult for you to interact with them”
M: manage cognitive dissonance	Patients and care partners often have difficulty accepting deprescribing recommendations because medications are supposed to make people healthier, not increase their risk of harm. To overcome this source of cognitive dissonance, clinicians can say, for example, “I am worried about your falls. Some of your medications may be hurting not helping you by clouding your mind and affecting your concentration and balance”
E: empower patients and care partners to continue the conversation.	Deprescribing conversations will often require multiple visits and should occur repeatedly over time—for example, immediately after dementia diagnosis and with each transition (i.e., from mild to moderate to advanced cognitive impairment). Patients and care partners should be encouraged to bring up deprescribing with all of their prescribers

care partners emphasized the importance of close follow-up from their clinicians throughout the deprescribing process, to offer guidance if symptoms recur after medications are stopped [43, 59]. Ensuring access to a physician or other member of the interdisciplinary team, such as a clinical pharmacist, was perceived as especially important for patients with cognitive impairment, who said that navigating automated clinic telephone menus could be confusing and make them fearful of deprescribing. Engaging interdisciplinary team members in this process can be very helpful. For example, clinical pharmacists can develop and monitor tapering plans, and nurses can check in with patients to ask about potential adverse drug withdrawal effects.

5 Strategies for Optimal Communication About Deprescribing

Good communication with patients and care partners is essential to engage patients and care partners in deprescribing, facilitate informed decisions, and ensure that medications are aligned with their goals of care. This is especially true for PWD, given their increased risk of iatrogenic harm and difficulty comprehending information and expressing their wishes as dementia progresses. In Table 3, we present communication strategies, adapted from the FRAME [60] tool, that are foundational to all aspects of the deprescribing framework for people with ADRD. These techniques can be especially useful in situations when patients or care partners are concerned about deprescribing or disagree with

a clinician's recommendation. In such instances, it can help to validate their experiences, link recommendations to their priorities (e.g., preventing falls or improving mental clarity), and reassure them that their symptoms will be addressed. Deprescribing is often a gradual process, requiring ongoing conversations over time.

6 Conclusions

Deprescribing is an essential component of high-quality care for PWD. Most clinical practice guidelines focus on single diseases and advocate initiating more medications. This is problematic for older adults with ADRD and multiple chronic conditions, such as Ms. R, for whom adding more and more therapies is burdensome and often harmful [61]. Yet few guidelines specifically address when and how to appropriately stop medications. For patients late in life, with multimorbidity, cognitive and functional impairment, the risk-benefit ratio for many medications is unclear and a patient's values, goals, and preferences should guide decisions about prescribing and deprescribing. Deprescribing for PWD in the primary care setting may require more in-depth conversations, perhaps facilitated by interdisciplinary team members, such as clinical pharmacists, to elicit patient and care partner goals and preferences, and inform them of the risks, benefits, and uncertainties of medications in the context of dementia.

A key lesson from trials, such as OPTIMIZE and DPREScribe-AD, is that deprescribing will require a cultural shift, in which *prescribing* is approached more judiciously, and patients and families are informed at the outset that every medication has tradeoffs and no medication should be prescribed forever. Changing health status and new diagnoses such as dementia may necessitate stopping medications that were once indicated. It is essential that all clinicians dedicate time to guide PWD, such as Ms. R and their care partners, through shared decision-making regarding medication use. The adapted framework and communication tools described herein can help clinicians provide patient-centered care, reduce the risk of iatrogenic harm, and improve well-being and quality of life for PWD.

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